

SYLLABUS

ESEC2213	ANALOG ELECTRONIC CIRCUITS	3-0-0	Credit-3
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Course Objectives:

Students will be able:

1. To get an overview of circuit realizations with diodes and transistors.
2. To get acquainted with signal analysis of BJTs and FETs and to have an overview of low and high frequency responses.
3. To understand the concept of feedback amplifier and oscillator.
4. To get an overview of power amplifier.
5. To learn operational amplifiers and its various applications.

MODULE-I

(12 Hours)

Small Signal Analysis of BJTs

Small Signal Analysis of CE amplifiers. Effects of R_s and R_L on CE amplifier operation, Emitter Follower, Cascade amplifier, Darlington Connection and Current Mirror Circuits.

Frequency Response of BJTs

Low and High Frequency Response of BJTs, Frequency Response of CE Amplifier, Multistage Frequency Effects, Miller Effect Capacitance, Square Wave Testing.

MODULE-II

(10 Hours)

Metal Oxide Semiconductor Field-Effect Transistor

Principle and Operation of MOSFETs, p-Channel and n-Channel MOSFET, Complimentary MOSFET, MOSFET as an Amplifier and as a Switch.

Small Signal Analysis of FETs

Small-signal model and biasing circuits, Common source amplifier, small signal equivalent circuits- gain, input and output impedance, transconductance, Effects of R_s and R_L on CS Amplifier, Source Follower and Cascaded System.

Frequency Response of FETs

Low and High Frequency Response of FETs, Frequency Response of CS Amplifier, Multistage Frequency Effects.

MODULE-III

(10 Hours)

Feedback amplifier and Oscillators

Concepts of negative and positive feedback, Four Basic Feedback Topologies, Practical Feedback Circuits, Principle of Sinusoidal Oscillator, Wein-Bridge, Phase Shift and Crystal Oscillator Circuits.

Power Amplifier

Classifications, Class-A, Class-B and Class-AB Amplifier Circuits, Transfer Characteristics, Power Dissipation and Conversion Efficiency of Power Amplifiers.

MODULE-IV

(10 Hours)

Operational Amplifier

Differential Amplifier, direct coupled multi-stage amplifier, internal structure of an Op-Amp. Non-idealities in Op-Amp (Output offset voltage, input bias current, input offset current, slew rate, gain bandwidth product). Open-loop and Closed-loop Gains.

Linear Applications of Op-Amp

Instrumentation amplifier, buffer circuit, current-to-voltage converter, voltage-to-current converter, active filter, P, PI and PID controllers, lead/lag compensator using an Op-Amp. Analog to digital conversion.

Non Linear Applications of Op-Amp

Logarithmic amplifier, Precision rectifier, clipper and clamper circuit, Op-Amp as comparator circuit, Peak detector, zero crossing detector, Square, triangular wave generator circuits using Op-Amps, Concept of Schmitt triggers circuit and sample and hold circuit using operational amplifier, 555 Timer and its applications.

Text Books:

1. Adel Sedra and Kenneth C Smith, "Microelectronics Circuits", Oxford University Press, 7th Edition, 2014.
2. Ramakant A. Gayakwad, "Op-Amps and Linear Integrated Circuits", Pearson Education, 4th Edition, 2013.
3. J. Milliman, C. Halkias, S. Jit, "Electronics Devices and Circuit", Tata McGraw Hill Education, 2nd Edition, 2008.
4. R. L. Boylestad and L. Nashelsky, "Electronic Devices and Circuits Theory", Pearson Education, 10th Edition, 2013.

Reference Books:

1. P. Horowitz, W. Hill, "The art of Electronics", Cambridge University Press, 1989.
2. Donald A. Neamen, "Electronics Circuits Analysis and Design", Tata McGraw Hill, 3rd Edition, 2002.
3. J. Milliman, C. Halkias, "Integrated Electronics: Analog and Digital Circuits and Systems", Tata McGraw Hill, 2nd Edition, 2004.
4. David A. Bel, "Electronic device and circuits", Oxford University Press, 5th Edition, 2008.
5. Anil K. Main, "Electronics devices and circuits", Wiley India Pvt. Ltd, 3rd Edition, 2009

Course Outcomes:

1. To design and analyze small and large signal amplifier circuits.
2. To analyze the characteristics of cascaded amplifiers.
3. To design feedback amplifiers and oscillators.

B.Tech Syllabus for Admission Batch 2018-19 Onwards

Computer Science & Engineering

4. To analyze the characteristics of operational amplifier in open and closed loop configurations.
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PYQs



Registration no:

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Total Number of Pages: 02

B.Tech.

Subject Code: ESEC2213

3rd Semester Regular/ Back Paper Examination January 2023

Subject Name: Analog Electronic Circuits

Branch: Computer Science & Engineering

Time: 3 Hours

Max Marks: 60

Question Code:

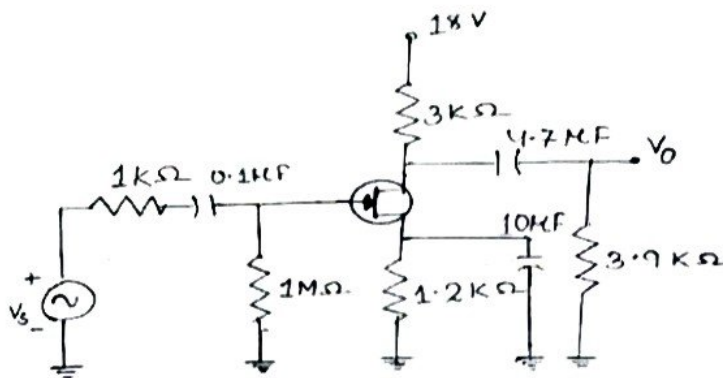
Answer Question No.1 which is compulsory and any five from the rest.

The figures in the right hand margin indicate marks.

Q1 Answer the following questions: (1x10)

- Why FET is called as "voltage operated device"?
- What is the Barkhausen criterion to sustain oscillation?
- Define CMRR of an op-amp.
- Write the expression of Fourier series expansion for a square wave.
- Draw the Darlington pair transistor circuit and define β_D .
- Define Threshold voltage in Mosfet.
- Prove that Negative feedback in an amplifier reduces gain.
- What is Unity gain bandwidth product.
- How does the bypass capacitor affecting to the Gain of amplifier?
- Draw the transfer characteristics curve of n channel E-Mosfet.

Q2 a) (5)



For the given circuit, $I_{DSS}=6\text{mA}$, $V_p=-6\text{V}$, $r_d=\infty$. Determine f_{LG} , f_{LC} , f_{LS} . What will be the lower cutoff frequency?.

b) Briefly explain the principle and operation of n channel D-MOSFET. Why is it called so? (5)

Q3 a) Find the input impedance, output impedance and Gain of voltage series feedback circuit. (5)

b) Derive the expression for finding f_L , f_H and bandwidth by square wave testing method. (5)



- Q4** a) Draw the circuit diagram of Phase shift oscillator and derive the expression for frequency of oscillation, Gain (A) and feedback gain (β). (7)
- b) Differentiate between negative feedback and positive feedback amplifier. (3)
- Q5** a) Draw the circuit of voltage divider bias(bypass). Take the circuit parameter as, $V_{CC}=10V$, $R_2=17K\Omega$, $R_1=83K\Omega$, $R_C=2K\Omega$, $R_E=0.5K\Omega$. The transistor has $\beta=100$ and $V_{BE}=0.7V$. Draw re model and find z_i , z_o , A_v . (5)
- b) Draw the equivalent two port network of above said circuit. If $R_s=0.6 K\Omega$, $R_L=3.3 K\Omega$ connected to the circuit then, what will be the overall voltage gain, input impedance, output impedance of the circuit? (5)
- Q6** a) Draw the circuit of class-A transformer coupled power amplifier and explain its operation. Derive an expression for maximum efficiency of conversion. (7)
- b) What is crossover Distortion ? Where is it found? How it can be overcome? (3)
- Q7** a) Draw the circuit diagram of an active low pass filter and explain its operation and find the cutoff frequency. (5)
- b) Draw the Instrumentation Amplifier circuit, derive the output voltage expression and find V_{out} .(when $V_1=3V$, $V_2=1V$ and the resistance of all the resistors is $5K\Omega$) (5)
- Q8** Write short notes on any two: (5x2)
- a) Miller capacitances (C_{Mi} and C_{Mo}).
 - b) Crystal oscillator
 - c) Current Mirror circuit
 - d) BJT Cascade connection

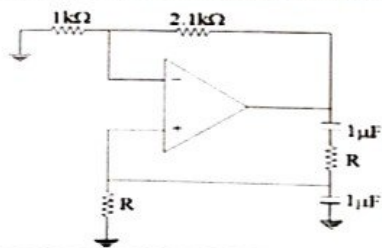
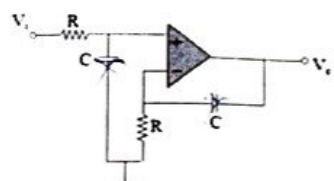


INDIRA GANDHI INSTITUTE OF TECHNOLOGY, SARANG
3rd Semester Regular/Back Paper Examination December 2023-24

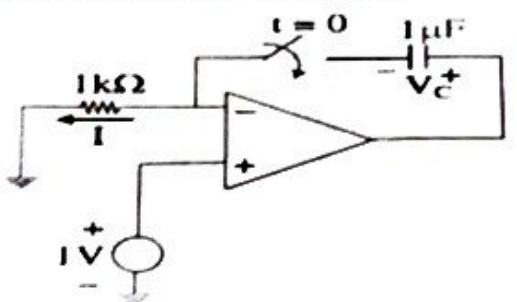
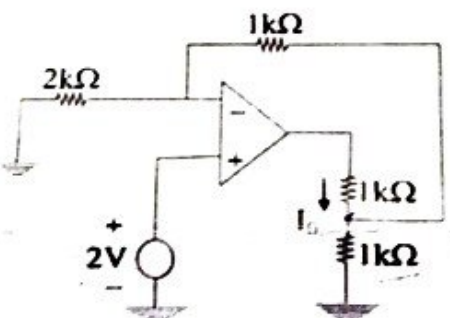
Total no. of Pages: 02

Branch	Computer Science Engineering	Programme	B. Tech
Course Name	ANALOG ELECTRONIC CIRCUITS	Semester	3 rd
Course Code	ESEC2213	Year	2023-24
Time: 03 Hrs	Answer Question number 1 which is compulsory and any five from the rest.	Maximum Marks	60

Each Question Carry Marks (M) as Indicated in the right-hand Margin

Q's	Questions	M	BL	CO	PO
1. a	What is an Amplifier? And what are the conditions required to satisfy to perform as a signal Amplifier?	01	L1	CO2	PO1
b	Draw a Trans-Resistance Amplifier? Justify your design?	01	L2	CO2	PO1
c	What is the input resistance of a simple Common emitter BJT?	01	L1	CO2	PO1
d	Explain Early effect on a BJT transistor?	01	L1	CO1	PO1
e	What are the steps required to find AC equivalent circuit?	01	L1	CO2	PO2
f	Is JFET is a current controlled device or voltage-controlled device, Justify?	01	L2	CO1	PO2
g	What is the input resistance of an OPAMP? and Why?	01	L1	CO5	PO1
h	What is virtual ground concept of an OPAMP?	01	L2	CO5	PO2
i	What is the Gain of an Inverting Amplifier?	01	L2	CO5	PO2
j	An oscillator requires what kind of feedback to operate?	01	L1	CO3	PO1
2. a	Find Current Gain, Voltage Gain and Input resistance(impedance) for CE Amplifier using h-parameter hybrid model? Explain the steps taken to avoid loading effect in the Amplifier design?	05	L2	CO2	PO1
b	Find the Current Gain, Voltage Gain, Input resistance, Output resistance of RC coupled single stage CE amplifier using Simplified analysis? Explain the use of Coupling and Bypass capacitor?	05	L2	CO2	PO1
3. a	Explain the working principle with neat diagram for a n-channel MOSFET?	05	L2	CO1	PO1
b	Find the Voltage Gain, Input Resistance and Output Resistance for a JFET common source Amplifier?	05	L2	CO2	PO1
4. a	Explain the working principle of a Push Pull Class B power amplifier and find its maximum efficiency?	05	L1	CO4	PO2
b	The value of 'R' required for the sinusoidal oscillations of frequency 1 KHz in the circuit of the figure? 	05	L4	CO3	PO3
5. a	Derive and explain the transfer function of a low pass filter with neat circuit diagram?	05	L6	CO2	PO1
b	If OPAMP shown in the below figure is ideal, Then what kind of filter it represents? 	05	L3	CO2	PO2

P.T.O

6.a	Derive the frequency of oscillation and condition for sustained oscillations for a WEIN BRIDGE OSCILLATOR?	05	L2	CO3	PO2
7.b	Draw and Explain different mixing and sampling network for a feedback circuit used in an amplifier?	05	L1	CO3	PO1
7.a	Explain the working of a SCHMITT TRIGGER circuit?	05	L3	CO5	PO2
8	Find V_C at $t=1\text{msec}$, If capacitor is initially uncharged?	05	L5	CO5	PO4
					
ANSWER ANY TWO QUESTIONS					
8.a	A common emitter transistor amplifier has the hybrid parameter of $h_{ie}=1100\Omega$, $h_{re}=2.5 \times 10^{-4}$, $h_{fe}=50$, $h_{oe}=25\mu\text{A/v}$ and $R_L=R_S=1000\Omega$. Find Current gain, Voltage Gain, Input resistance and Output resistance using Simplified Analysis?	05	L5	CO2	PO2
b	Draw a simplified h-parameter for a Common Base and derive its voltage gain, current gain, input and output resistance?	05	L2	CO2	PO2
c	Find the output voltage of a OPAMP based subtractor?	05	L6	CO5	PO3
d	Find the output current in figure shown below. Assume OPAMP to be ideal?	05	L5	CO5	PO4
					

Registration No.

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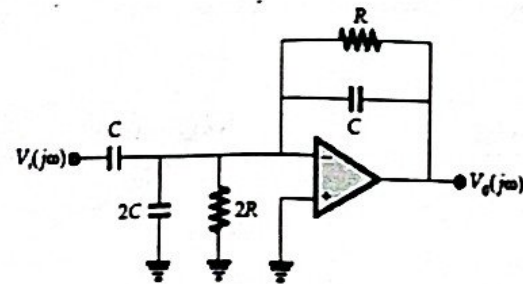
INDIRA GANDHI INSTITUTE OF TECHNOLOGY, SARANG
3rd Semester Regular/ Back Paper Examination AY 2024-25

Total no. of Pages:
02

Branch	Computer Science & Engineering	Programme	B. Tech
Subject Name	ANALOG ELECTRONIC CIRCUITS	Semester	3 rd
Subject Code	ESEC2213	Year	2024-25
Time: 03 Hrs	Answer Question number 1 which is compulsory and any five from the rest.	Maximum Marks	60

Each Question Carry Marks(M) as Indicated in the right-hand Margin

Q's	Questions	M	BL	CO	PO
1.a	What is an Oscillator? And what are the conditions required to satisfy to perform as an signal Oscillator?	01	L1	CO2	PO1
b	Draw a Voltage Amplifier? Justify your design?	01	L1	CO2	PO1
c	What is the input resistance of a simple common emitter BJT?	01	L1	CO2	PO1
d	Explain the frequency response of an amplifier?	01	L1	CO1	PO1
e	What is the significance of bypass capacitor in a common emitter BJT?	01	L1	CO2	PO2
f	JFET works in depletion mode, Justify?	01	L2	CO1	PO2
g	What is common mode voltage gain and differential voltage gain in an OPAMP?	01	L1	CO5	PO1
h	What is virtual ground concept of an OPAMP?	01	L2	CO5	PO2
i	What is a voltage follower? Justify	01	L2	CO5	PO2
j	Why an amplifier is incorporated with an oscillator circuit?	01	L1	CO3	PO1
2.a	Find Current Gain, Voltage Gain and Input resistance(impedance) for CE Amplifier using h-parameter hybrid model? Explain the steps taken to avoid loading effect in the Amplifier design?	05	L2	CO2	PO1
b	Find the Current Gain, Voltage Gain, Input resistance, Output resistance of RC coupled single stage CE amplifier using Simplified analysis? Explain the use of Coupling and Bypass capacitor?	05	L2	CO2	PO1
3.a	Derive the small signal model for a JFET Amplifier?	05	L2	CO1	PO1
b	Find the Voltage Gain, Input Resistance and Output Resistance for a JFET common source Amplifier?	05	L2	CO2	PO1
4.a	Explain the working principle of a Push Pull Class B power amplifier and find its maximum efficiency?	05	L1	CO4	PO2
b	In a current mirror given below derive and find I_{REF} ?	05	L4	CO3	PO3
5.a	Derive and explain the low frequency response of a BJT Amplifier?	05	L3	CO2	PO1
b	If OPAMP shown in the figure is ideal, Then what kind of filter it represents? If $R=2K\Omega$ and $C=1\mu F$ then what is the cutoff frequency?	05	L6	CO2	PO2



$\omega = \frac{1}{RC}$

6.a

Draw and Explain different feedback topology circuit used in an amplifier?

05

L2

CO3

b

Derive and Explain the frequency response for a Low pass filter?

05

L1

CO3

7.a

Derive the frequency of oscillation and condition for sustained oscillations for a RC PHASE SHIFT OSCILLATOR?

05

L3

CO5

b

For an unknown transistor oscillator $L=100 \mu H$, $C_1=0.005 \mu F$ and $C_2=0.01 \mu F$, Identify the oscillator and find the frequency of oscillation generated in KHz?

05

L5

CO5

Short notes type Questions Only (Answer any Two)

8.a

A common emitter transistor amplifier has the hybrid parameter of $h_{ie}=1100 \Omega$, $h_{re}=2.5 \times 10^{-4}$, $h_{fe}=50$, $h_{oe}=25 \mu A/V$ and $R_L=R_S=1000 \Omega$. Find Current gain, Voltage Gain, Input resistance and Output resistance using Simplified Analysis?

05

L2

CO2

b

Explain the working of a square wave generator?

05

L6

CO5

c

Find the output voltage of an OPAMP based adder?

05

L5

CO5

d

Find the output voltage V_{out} in figure shown below. Assume OPAMP to be ideal?

05

L4

CO2

